Generative AI Models and Architecture

What Makes Generative AI Possible?

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Source: https://commons.wikimedia.org/wiki/File:Kasparov_Magath_1985_Hamburg-2.png

Artificial Intelligence

Enabling computers to mimic human behavior

Old AI = logic and rules

New AI = learning and generation

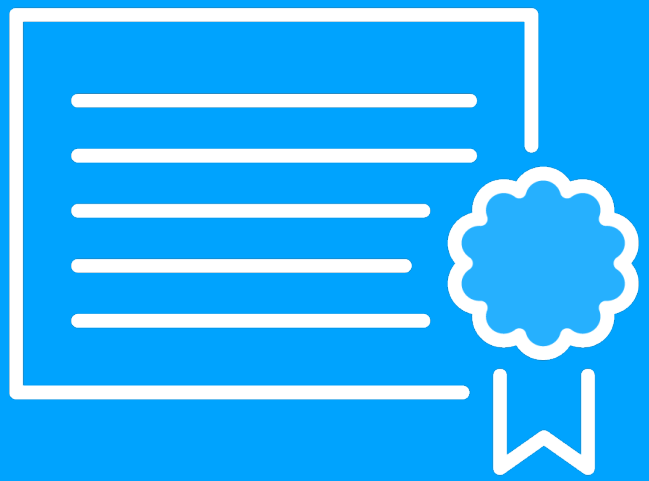
New Architectures and Strategies

Transformers

Generative adversarial networks

Variational autoencoders

Diffusion models



Prerequisites

Familiarity with generative AI tools like ChatGPT

How AI Reads: From Words to Tokens

Generative AI is changing everything

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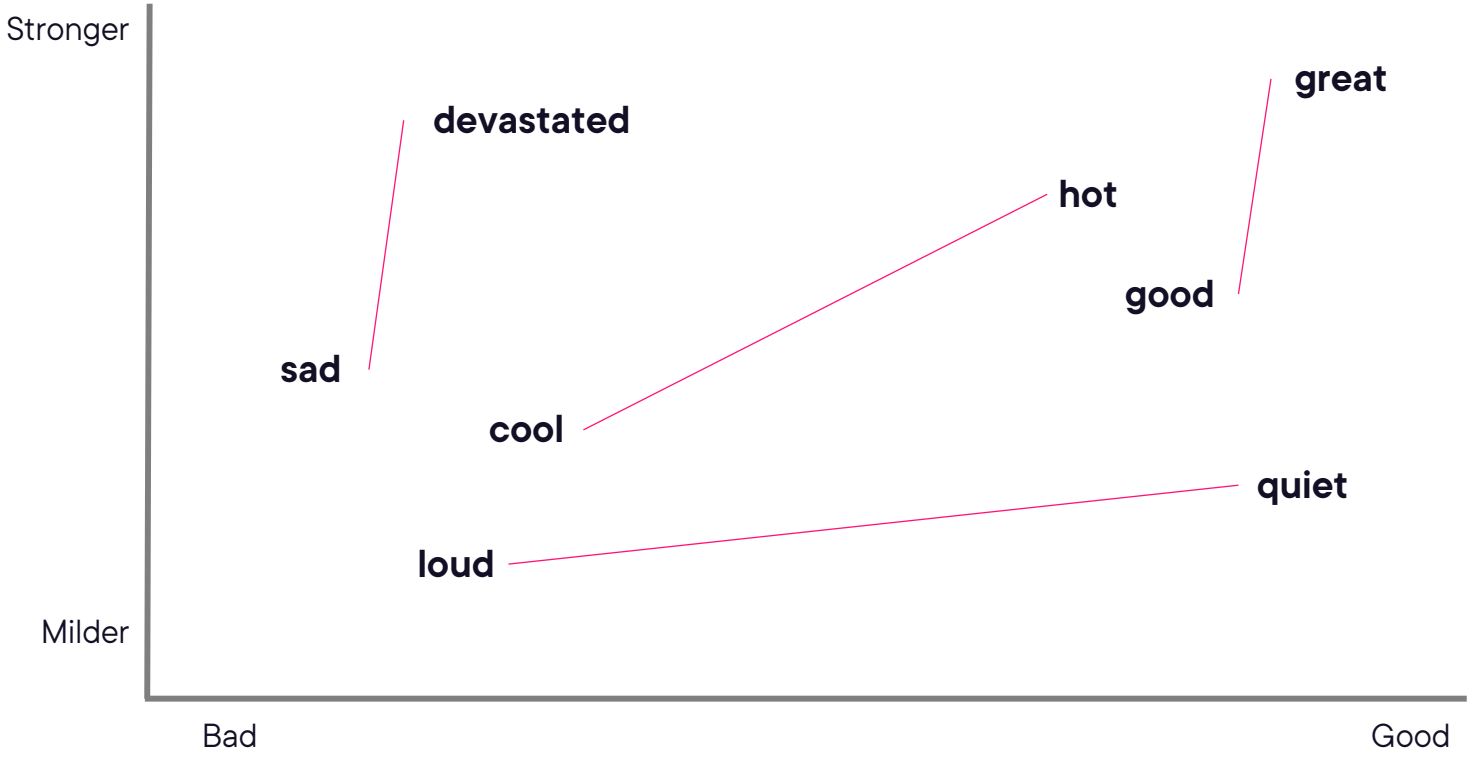
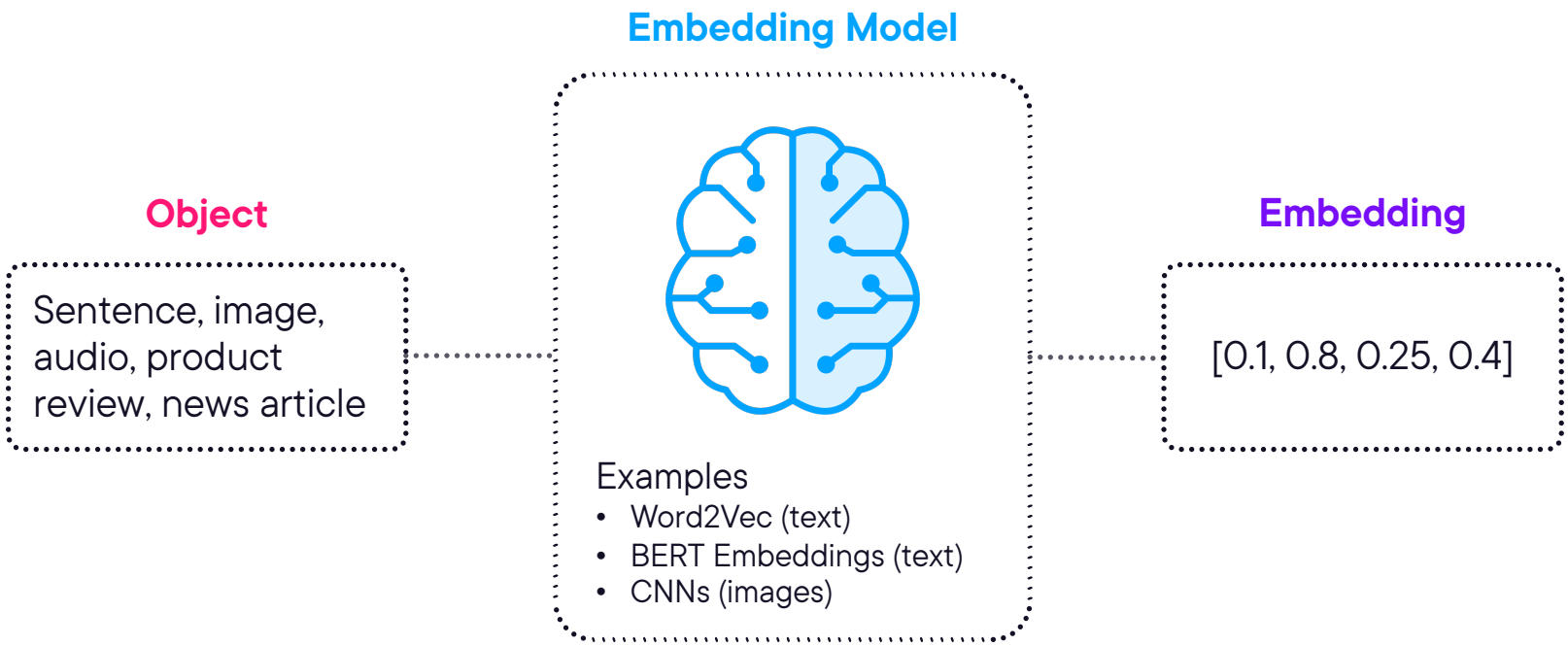
What Are Embeddings?


Stronger
Milder
Bad
Good
great
devastated
hot
good
sad
cool
quiet
loud

Vectors and Embeddings

Vector	vs.	Embedding
A list of numbers that can describe anything		A vector with learned meaning that represents complex concepts
Fixed or manually assigned		Learned by AI models
No inherent relationships		Captures similarities and context
Example: A coffee shop in NYC		*Example: My favorite coffee shop in NYC*
[40.7128, -74.0060]		*[0.82, 0.45, 0.67, 0.91, 0.33]*
Latitude and longitude of the coffee shop		*Captures additional qualities like vibe, coffee quality, popularity, proximity to other favorite spots*


Embedding Model
Object
Sentence, image, audio, product review, news article
Examples
• Word2Vec (text)
• BERT Embeddings (text)
• CNNs (images)
Embedding
[0.1, 0.8, 0.25, 0.4]

Recapping

1. Start with text (like your prompt)
2. Break it into tokens (tokenization)
3. Map tokens to token IDs (labels)
4. Convert token IDs to vectors (a list of numbers)
5. Use vector embeddings to understand relationships and meaning

Attention Is All You Need

Attention/Self-attention

The cat chased the mouse

Attention/Self-attention

The mouse controls the cursor

Attention/Self-attention

The cat sat on the mat because it was tired

Positional Encoding

Pulling It All Together

Understands relationship between words and the overall context

The cat chased the mouse

The Story of the Cat and the Mouse

In a quiet little village nestled between rolling green hills, there lived a sleek, gray cat named Whiskers. Whiskers was known throughout the village as a skilled hunter. No mouse dared venture near the cottages when Whiskers was prowling about — except for one.

That mouse's name was Pip, a tiny, clever creature with a patch of white fur shaped like a heart on his chest. Pip wasn't like other mice. Instead of sneaking into pantries for crumbs, he loved to explore and collect shiny objects. One moonlit night, Pip's curiosity led him into trouble.

While searching for a gleaming button he had spotted earlier, Pip scurried into a cottage — the very same one where Whiskers lived. As Pip reached for the button, a shadow loomed over him.
...

Works for large amounts of text too

The Transformer Architecture

The cat sat on the mat.
Transformer
Le chat s’est assis sur le tapis.

1 **Input**

The cat sat on the mat.

Embeddings and positional encoding

2 Encoder

Understand the input

- Multi-head attention
- Feedforward network
- Add and norm

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Context-rich vectors

1 Input

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Embeddings and positional encoding

3 Generation Starts

Generate the output (one token at a time)

- Start with beginning of sequence <BOS>
- Shift outputs from earlier steps <BOS> Le chat...

2 Encoder

Understand the input

- Multi-head attention
- Feedforward network
- Add and norm

4 Decoder Stack

Generate next word (using attention and context)

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Generate the output (one token at a time)

- Start with beginning of sequence <BOS>
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5 Linear + Softmax

Convert decoder output into word probabilities